5→ Laws of chemical combination

- **Law of conservation of mass** :- (Lavoisier) → matter can neither be created nor destroyed
  - → In a chemical reaction total mass remains conserved.
  - → Mass before reaction = Mass after reaction.

eg $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$

Mass before rxn: $2\times1 + \frac{1}{2}\times32$ → 18 gm

mass after rxn: $2\times1+16$ → 18 gm

Exception → Nuclear rxn

- **Law of definite proportion** (proust) → all chemical compounds have constant composition by weight, irrespective of method of preparation or source.

eg. Cupric carbonate	% of Cu	% of oxygen	% of carbon
natural sample	51.35	9.74	38.91
Synthetic "	51.35	9.74	38.91

Que 1.8g of a metal burnt in oxygen & formed 3gm oxide. 1.5gm of same metal burnt in oxygen to form 2.5 gm oxide. check if they follow law of definite propoth

⇒ Sample1

M − 1.8g

Oxide − 3gm

oxygen → (3 − 1.8) → 1.2

$\frac{M}{\text{oxygen}} = \frac{1.8}{1.2} = 1.5$

Sample2

M = 1.5g

Oxide = 2.5g

oxygen = 1gm

$\frac{M}{\text{oxygen}} = \frac{1.5}{1} = 1.5$

yes it follows law of definite proportion

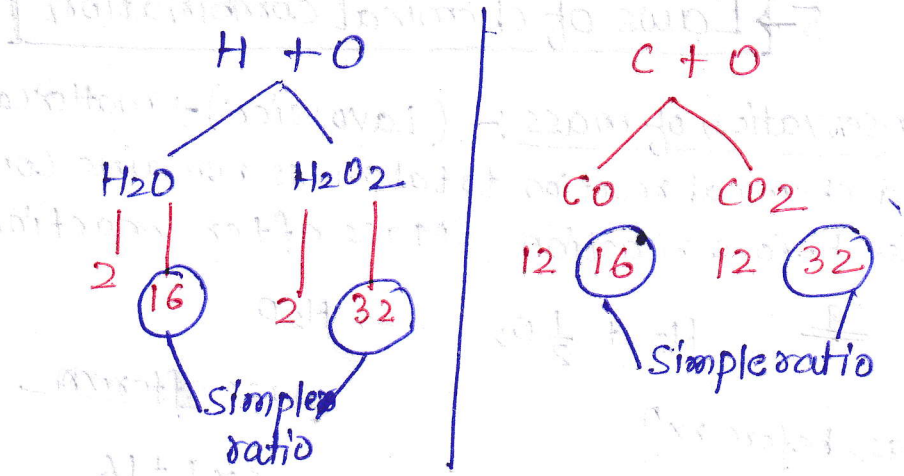
Exception $^{12}CO_2$ $\frac{12}{32}$ $^{14}CO_2$ $\frac{14}{32}$ } fail.

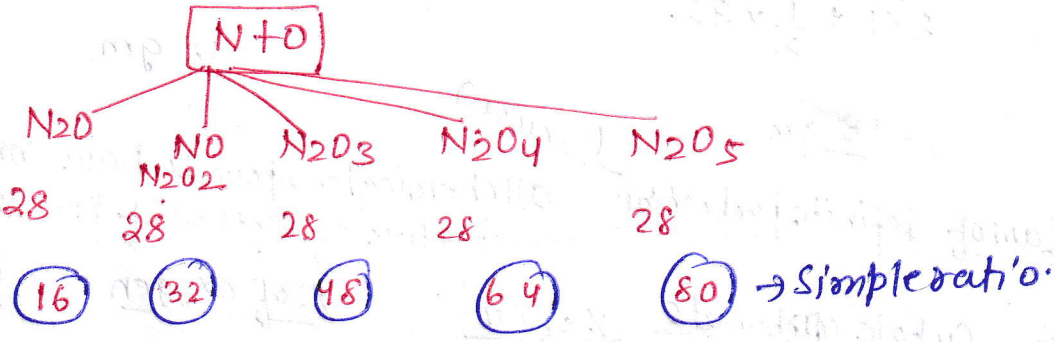
- **Law of multiple proportion** (Dalton) : when two elements combine to give two or more product, the mass of one element which combines with const mass of other element are in simple ratio.

12gm C + $\frac{1}{2}O_2$ (16gm) → CO

12gm C + O_2 (32gm) → CO_2

if mass of Carbon is fixed : oxygen mass in simple ratio 16 : 32 = 1 : 2





• Gay Lussac's law → Gases combine a reaction by simple ratio of volume. (given temp & pressure are const)
Scientist

Ex $3H_{2(g)} + N_{2(g)} \longrightarrow 2NH_{3(g)}$

(For gas mole = vol)

ratio of volume = 3 : 1 : 2

Ex $2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(g)}$

100 ml 50 ml 100 ml

ratio of vol = 2 : 1 : 2

• Avogadro law :- at same temp & pressure equal vol of gas contains equal no of molecules & not atoms.

	1 litre	1 litre	1 litre
	He	N_2	O_3
no of molecule →	1	1	1
no of atom →	1	2,	3

Dalton's atomic theory → Atomic str.

Molecular mass → Sum of atomic mass of elements present in a molecule

RMM

O_2 = 32

O_3 = 48

H_2SO_4 = 98

mm

$O_2 = 2 \times 16\ amu = 32\ amu$

$O_3 = 3 \times 16\ amu = 48\ amu$

$HCl = (1 + 35.5)\ amu = 36.5\ amu$

$H_2SO_4 = 98\ amu$

Gmm (molarmass)

$O_2 \rightarrow 32\ amu \times N_A = 32\ g$

$O_3 \rightarrow 48\ gm$

$HCl \rightarrow 36.5\ gm$

$H_2SO_4 \rightarrow 98\ gm$

Que what is molecular mass of N_2 ?? (in gm)
what is gram molecular mass of N_2 ?? (in gm)

→ 28 amu

$28 \times 1.66 \times 10^{-24}$ gm

46.48×10^{-24} gm

4.648×10^{-23} gm

$28\ amu \times N_A$

$28\ gm$

Que How many atoms of oxygen are there in 32 gram oxygen ?

~~$16\ amu \times N_A = 16\ gm$~~

mass of 1 atom = 16 amu

$16\ amu \times x\ atom = 32\ gm$

$$x = \frac{32}{16\ amu}$$

$$x = 2 \times \frac{1}{amu}$$

$1\ amu \times N_A = 1$

$x = 2N_A$

Avg atomic mass

Type-1 if isotopes are in %

$$\text{av atomic mass} = \frac{a_1x_1 + a_2x_2 + a_3x_3 + \ldots + a_nx_n}{100}$$

$a_1 a_2 a_3 \ldots$ atomic mass

$x_1 x_2 x_3 \ldots$ % of isotopes.

Que Chlorine has 75% Cl^{35}, ~~atomic mass of 35 amu~~ and 25% Cl^{37}.
Cal av. atomic mass.

$$\text{av atomic mass} = \frac{35 \times 75 + 37 \times 25}{100} = 35.5\ amu$$

Type-2 if isotopes are in fraction → $\text{av mass} = a_1f_1 + a_2f_2 + a_3f_3 + \ldots + a_nf_n$

- **RAM (Relative atomic mass)**

$$RAM = \frac{\text{mass of one atom of an element}}{\frac{1}{12} \times \text{mass of an atom of Carbon (C-12)}}$$

→ no unit (RAM = no of nucleons)

- **Atomic mass unit (amu) (u)**

$$1\ amu = \frac{1}{12} \times \text{mass of one atom of C-12}$$

$$1\ amu = 1.66 \times 10^{-24}\ g$$
$$= 1.66 \times 10^{-27}\ kg$$

- **Atomic mass** : mass of 1 atom in amu.

$$AM = RAM \times 1\ amu$$
$$MM = RMM \times 1\ amu$$

Que Find relative atomic mass of oxygen atom & its atomic mass in gram.

→ RAM = 16

AM = RAM × 1 amu

AM = 16 amu

$= 16 \times 1.6 \times 10^{-24}$ gm

$= 25.6 \times 10^{-24}$ gm

$= 2.56 \times 10^{-23}$ gm

Gram atomic mass → mass of 6.02×10^{23} atoms
or mass of 1 mole atom
or molar mass.

RAM of 'O' = 16

AM of 'O' = 16 amu

GAM of 'O' = $16\ amu \times 6.02 \times 10^{23}$

$= \frac{16 \times 1.6 \times 10^{-24} \times 6.02 \times 10^{23}}{1}$ gm

$= 16$ gm

$1\ amu \times N_A = 1$

	RAM	AM	GAM
H	1	1amu	1gm
He	4	4amu	4gm
C	12	12amu	12gm
N	14	14amu	14gm

Que $Cl^{35} \rightarrow 0.75$ + $Cl^{37} \rightarrow 0.25$

av. mass = $35 \times 0.75 + 37.025$

= 35.5 amu

Que X^{200} (90%), X^{201} (8%), X^{202} (2%). Find av. mass of X.

$$\text{av mass} = \frac{200\times90 + 201\times8 + 202\times2}{100} = 200$$

$$= \frac{2(200\times45 + 201\times4 + 202\times1)}{100}$$

$$= \frac{9000 + 802 + 202}{50}$$

$$= \frac{10004}{50} \simeq 200 \text{ amu}$$

Average molar mass

Que

	at wt	% composition
oxygen	16	16%
nitrogen	14	80%
CO_2	—	3%
SO_2	—	1%

⇒ Find av molecular mass of mixture.

$$\text{av. molar mass} = \frac{m_1 \cdot x_1 + m_2 x_2 + \dots + m_n x_n}{100}$$

$$= \frac{16\times32 + 80\times28 + 3\times44 + 1\times64}{100}$$

$$= \frac{2948}{100} = 29.48$$

Formula mass → For ionic compounds, it represents ratio of different ions.

eg:- NaCl → In NaCl, one Na^+ is surrounded by 6 Cl^- and one Cl^- is surrounded by 6 Na^+.

↓

Formula mass = 23 + 35.5

= 58.5 u

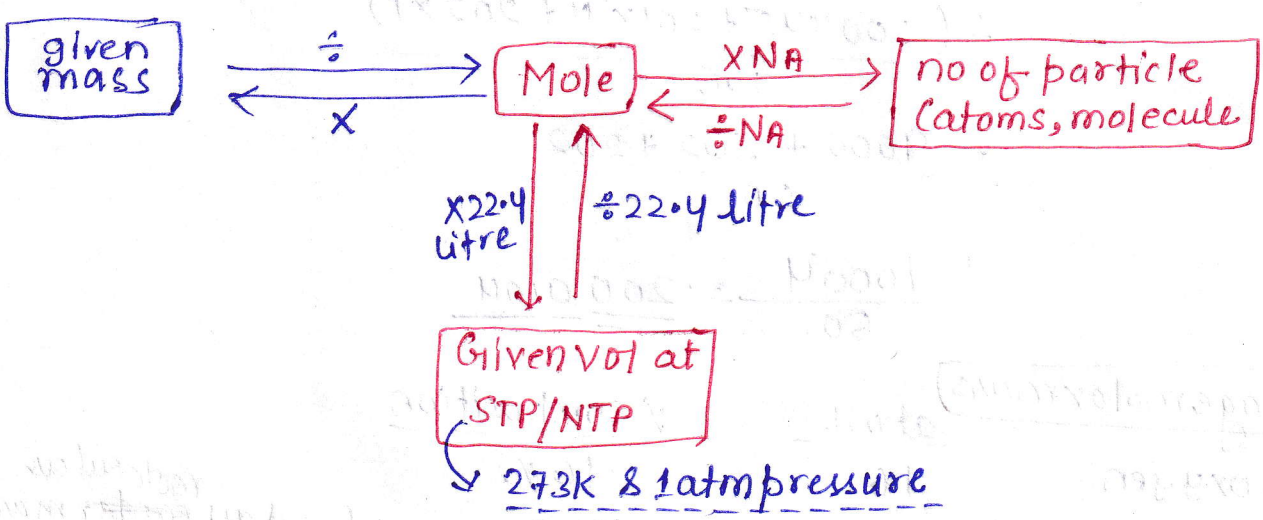
Mole

1 dozen = 12
1 Hundred = 100
1 mole = 6.02×10^{23}
= NA

- 1 mole atom = 6.02×10^{23} atom
- 1 mole molecule = 6.02×10^{23} molecule
- 1 mole ion = 6.02×10^{23} ion
- 1 mole rupee = 6.02×10^{23} rupee

mass of 1 mole atom = gram atomic mass
" " " molecule = gram molecular mass } molar mass

Formula



Que Cal no of atoms, molecules in 1 gm Nitrogen 22.

$$\text{mole} = \frac{1}{28}$$

1 N_2 ⟶ 2 atom

1 mole N_2 ⟶ 2 mole atom

$\frac{1}{28}$ mole N_2 ⟶ $2 \times \frac{1}{28}$ ~~atom~~ mole atom

⟶ $\frac{1}{14}$ mole atom

⟶ $\frac{1}{14} \times N_A$ atom

~~1 mole ⟶ 1 molecule~~

1 mole ⟶ N_A molecule

$\frac{1}{28}$ " ⟶ $\frac{N_A}{28}$ "

Que Cal Vol in litre of 20 gram hydrogen gas at STP.

$$\text{mole} = \frac{20}{2}$$

$\boxed{n = 10}$

1 mole – 22.4 litre

10 mole ⟶ 224 litre

Que Find mass of 10^{21} ~~molecules~~ atoms of Cu. (at wt in gm = 63.5)

$$\text{mole} = \frac{10^{21}}{6.02 \times 10^{23}} = \frac{1}{600} \text{ mole}$$

1 mole Cu → 63.5 gm

$\frac{1}{600}$ mole Cu → $\frac{63.5}{600}$ gm

Que Cal no of atoms of chlorine in 2.08 gm $BaCl_2$ {Ba – 137 gm, Cl – 35.5 gm}

mole = $\frac{2.08}{208}$ = 0.01 mol

1 $BaCl_2$ → 2 Cl atom

1 mole $BaCl_2$ → 2 mol Cl atom

0.01 mole $BaCl_2$ → 0.02 mol Cl atom

→ 0.02 N_A

Que Gn ½ gm molecule of oxygen.

Find mass, no of molecule, vol at STP, no of oxygen atom

⇒ mole = gm molecule = $\frac{1}{2}$

mole = $\frac{\text{mass}}{\text{molar mass}}$

mass = $\frac{1}{2} \times 32 = 16$ gm

~~1 mole →~~

~~O_2 →~~

1 mole → ~~1 mole~~ N_A molecule

$\frac{1}{2}$ mol → $\frac{N_A}{2}$,,

1 mol – 22.4 l

$\frac{1}{2}$ mol – 11.2 litre

1 O_2 → 2 atom

1 mole → 2 N_A

$\frac{1}{2}$ mole → N_A atom

Que Cal no of atoms, molecules in 1.2 gm ozone.

$n = \frac{1.2}{48}$

$n = \frac{1}{40}$

1 O_3 → 3 atom

1 mol O_3 → 3 mole atom

$\frac{1}{40}$ mole → $\frac{3}{40}$ mole = $\frac{3N_A}{40}$ atom

1 mole – N_A molecule

$\frac{1}{40}$,, ⇒ $\frac{N_A}{40}$,,

Que Cal no of atom in one drop of water having mass 1.8gm

$\Rightarrow$ $n = \frac{1.8}{18} = 0.1$ mole

1 H_2O → 3 atom

1 mol " → 3 mol atom

0.1 mol " → 0.3 mol atom

→ $0.3N_A$

Que Cal mass of one atom of Nitrogen in gram.

14 amu

$14 \times 1.6 \times 10^{-24}$ gm

22.4×10^{-24} gm

Que Calculate mass of one molecule of H_2O in gram.

18 amu

$18 \times 1.6 \times 10^{-24}$ gm

28.8×10^{-24} gm

Que Cal no of atom in

(i) 1 mole N_2 → 1 N_2 → 2 atom → 1 mole N_2 → 2 mole atom / $2N_A$

(ii) 1 mole P_4 → $4N_A$

(iii) 0.05g H_2O →

↳ $n = \frac{0.05}{18}$

$= 0.002$

1 H_2O → 3 atom

1 mole H_2O → 3 mole atom

0.002 mole H_2O → 0.006 mol atom

→ $0.006N_A$

Percentage composition

$$\text{mass \% of an element} = \frac{\text{mass of that element in 1 mole comp}}{\text{molar mass}} \times 100$$

eg

mass % of H in H_2O $= \left(\frac{2gm}{18gm} \times 100\right)\% = 11.18\%$.

" " of O " " $= \left(\frac{16gm}{18gm} \times 100\right)\% = 88.79\%$.

Eg

mass % of C in $C_2H_5OH = \left(\frac{24}{46} \times 100\right)\%$.

H $= \left(\frac{6}{46} \times 100\right)\%$.

O $= \left(\frac{16}{46} \times 100\right)\%$.

Eg mass % of S in $H_2SO_4 = \left(\frac{32}{98} \times 100\right)\%$.

O $= \left(\frac{64}{98} \times 100\right)\%$.

H $= \left(\frac{2}{98} \times 100\right)\%$.

Empirical & molecular formula

molecular formula → Formula having actual num of atoms in a molecule.

Empirical " → Simplest whole number ratio of atom in a molecule.

	M.F	E.F
Glucose →	$C_6H_{12}O_6$	CH_2O

$$MF = (E.F) \times n$$

$$n = \frac{\text{molecular formula mass}}{\text{Empirical formula mass}}.$$

Que acetylene & benzene both have empirical formula CH. The molecular mass of acetylene & Benzene are 26 & 78. Find M.F.

⇒ ~~mf mass = 7~~.

Empirical formula = CH

" " mass = 13

acetylene

$n = \frac{26}{13} = 2$

$MF = (CH) \times 2$

$= C_2H_2$

Benzene

$n = \frac{78}{13} = 6$

$MF = (EF) \times 6$

$= C_6H_6$

Que An organic compo containing C, H, O has composition as follows:-

C → 40.687%, H = 5.085%, O = 54.228%.

The molecular wt of compd is 118. Cal mf of comp.

⇒

% of element	atmass	Relative no of atom	Simplest ratio	Simple whole no
C → 40.687	12	40.687/12 = 3.39	3.39/3.389 = 1	1
H — 5.085	1	5.085/1 = 5.085	5.085/3.389 = 1.5	2
O — 54.228	16	54.228/16 = 3.389	3.389/3.389 = 1	1

Ef

$C_2H_3O_2$

Ef mass = 59

$n = \frac{118}{59} \Rightarrow 2$

$MF = C_4H_6O_4$

Que a compd contains 4.07% H, 24.27% C, 71.6% Cl. Molar mass of compd is 98.96 gram. What is E.F & M.F.

⇒

	%	at wt	% / at wt		
H	4.07	1	4.07/1 = 4.07	4.07/2.021	2
C	24.27	12	24.27/12 = 2.021	1	1
Cl	71.6	35.5	71.6/35.5 = 2.021	1	1

Ef (CH_2Cl) Ef mass = 49.5

$n = \frac{98.96}{49.5}$

$n = 2$ $\quad$ mf = $C_2H_4Cl_2$

Stoichiometry & stoichiometric Calculation

applicable on balanced chemical reaction

Eg1

$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

1mole	2mol	1mol	1mol	1mol / 22.4 litre at STP
100gm	73gm	111gm	18gm	44gm

Eg2

$N_2 + 3H_2 \longrightarrow 2NH_3$

1mol	3mol	2mol
28gm	6gm	34gm
22.4 l	3x22.4 l	2x22.4 l

for gases $\Rightarrow$ mole = vol (no of mol = no of mol)

1 litre	3 litre	2 litre.
1000 ml	3000 ml	2000 ml

acc to avogadro law :- (volume = no of molecule = no of mol)

1 molecule	3 molecule	2 molecule

Mass - mass relation

Mass - Vol relation

Vol - Vol relation

mass-mass rel^n

Qu How much iron is obtained in the reduction of 1kg Fe_2O_3.

$Fe_2O_3 + H_2 \longrightarrow Fe + H_2O$

$\Rightarrow$ Step 1 $\rightarrow$ Balance the rx^n

$Fe_2O_3 + 3H_2 \longrightarrow 2Fe + 3H_2O$

1mol $\qquad$ 2mol

$\rightarrow 6.25$

mole of $Fe_2O_3 = \frac{1000}{160} = 6.25$ mole

(2×6.25) mol $\Rightarrow$ 12.5 mol

mass = 12.5 x 56 = 700 gram

Que 367.5 gm $KClO_3$ when heated. How many gram of KCl and Oxygen is produced (molar mass of $KClO_3$ = 122.5 gm)

→

$$KClO_3 \longrightarrow KCl + \frac{3}{2}O_2$$

$n = \frac{367.5}{122.5}$

$n = 3$

$KClO_3$	KCl	$\frac{3}{2}O_2$
1 mol	1 mol	$\frac{3}{2}$ mol
3 mol	3 mol	$\frac{9}{2}$ mol
	↓	
	3×74.5 gm	$\frac{9}{2} \times 32$
	223.5 gm	144 gm

Que what gram of AgCl is formed by rxn of 5.85 g NaCl on excess of $AgNO_3$.

$$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$$

$n = \frac{5.85}{58.5}$

$n = 0.1$

NaCl	$AgNO_3$	AgCl	$NaNO_3$
1 mol		1 mol	
0.1 mol		0.1 mol	
		0.1×143.5 gm	
		14.35 gm	

Mass-Volume Relation

Que at 100°C, for complete combustion of 3 gm ethane the req vol of O_2 & produced vol of CO_2 at STP will be??

⇒

$$C_2H_6 + \frac{7}{2}O_2 \longrightarrow 2CO_2 + 3H_2O$$

$n = \frac{3}{30}$

$n = 0.1$

C_2H_6	$\frac{7}{2}O_2$	$2CO_2$	$3H_2O$
1 mol	$\frac{7}{2}$ mol req	2 mol produce	
0.1 mol	$\frac{0.7}{2}$ mol req	0.2 mol produce	
	0.35 mol	0.2 mol	
	0.35×22.4	0.2×22.4	
	7.84 litre	4.48 litre	

Que In the rxn, if 10gm H_2 react with N_2, what will be the vol of NH_3 at STP.

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

3 mol → 2 mol

$n_{H_2} = \frac{10}{2}$

1 mol → $\frac{2}{3}$ "

$n = 5$

5 mol → $\frac{10}{3}$ mol

$\frac{10}{3} \times 22.4$ litre

74.64 litre

Vol–Vol Relⁿ

Que at 100°C, for complete combustion of 1.12 litre butane (C_4H_{10}) the produced vol of $H_2O(g)$ & $CO_2(g)$ at STP will be :-

→ $C_4H_{10} + \frac{13}{2}O_2 \longrightarrow 4CO_2(g) + 5H_2O(g)$

$n = \frac{1.12}{22.4}$ lit

1 mol → 4 mol → 5 mol

$\frac{1}{20}$ " → $\frac{4}{20}$ mol → $\frac{5}{20}$ mol

$n = \frac{1}{20}$

$\frac{1}{5}$ mol → $\frac{1}{4}$ mol

$\frac{1}{5} \times 22.4$ → $\frac{1}{4} \times 22.4$

$= 4.48$ l → $= 5.6$ l

Que 3 litre of a mixture of Propane & butane on complete combustion gives 10 litre CO_2. Find composition of mixture.

→ Propane ⇒ x litre ; butane → (3−x) litre.

x litre $C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$

$3x$

(3−x) litre $C_4H_{10} + \frac{13}{2}O_2 \longrightarrow 4CO_2 + 5H_2O$

$4(3-x)$

given :- $3x + 4(3-x) = 10$

$3x + 12 - 4x = 10$ ⇒ $x = 2$

$C_3H_8 \rightarrow 2$ litre

$C_4H_{10} \rightarrow 1$ litre.

Balance Rxn | Limiting Reagent | (when more than reactant's quantity is given)

- The reactant which is completely consumed in a rxn.

$$A + 5B \longrightarrow C + 3D$$

Given: 10 mol, 10 mol

For LR, divide by Stoichiometry: $\frac{10}{1}$, $\frac{10}{5}$

10, (2)

↓ least value shows L.R → Then apply stoichiometry on LR to solve question.

Que 6 mol of Na_2CO_3 is reacted with 4 mol of HCl solution. Find vol of CO_2 gas produced at STP.

$$Na_2CO_3 + HCl \longrightarrow NaCl + CO_2 + H_2O$$

Balance rxn ⇒

$$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$$

6 mol, 4 mol

L.R → $\frac{6}{1}$, $\frac{4}{2}$

6, 2

L.R

2 mol HCl → 1 mol CO_2

1 mol " → $\frac{1}{2}$ " "

4 " " → 2 " "

2×22.4 l

44.8 l

Que 50 kg N_2 & 10 kg H_2 are mixed to form NH_3. Cal NH_3 produced. Identify L.R in rxn.

$(n)_{N_2} \rightarrow \frac{50 \times 10^3}{28} = \frac{1785 \times 10^3}{1785}$

$(n)_{H_2} = \frac{10 \times 10^3}{2} = 5000$

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

1785, 5000

1785, $\frac{5000}{3}$

1667 → LR

3 mol H_2 → 2 mol NH_3

1 mol " → $\frac{2}{3}$

5000 → $\frac{10000}{3}$ mol

mass = $\left(\frac{10000}{3} \times 17\right)$ gm = $\frac{170000}{3}$

= 56666 gm = 56.66 kg

Reaction in solution

1. mass %
2. mole fraction
3. molarity
4. molality
5. ~~Normality~~
6. ~~Equivalent mass~~

1, 2 → independent of Temp

3 → dependent on temp. $\left(M \propto \frac{1}{T} \propto \frac{1}{Vol}\right)$

4 → Independent of Temp.

mass % (% w/w)

$$\text{mass \%} = \frac{\text{mass of solute}}{\text{mass of solution (solute + solvent)}} \times 100$$

Que A solution is prepared by adding 5 gm of substance X to 18 gm H_2O. Cal many. of solute.

$$\text{many \%} = \frac{5}{(18+5)} \times 100 = 21.74\%$$

Que 0.5 g of a solute is dissolved in 25 gm solvent. cal many % of solute

$$\text{many \%} = \frac{0.5}{25.5} \times 100 = 1.96\%$$

mole fraction (X)

$$X_A = \frac{n_A}{n_A + n_B}$$

$$X_B = \frac{n_B}{n_A + n_B}$$

Que A solution is prepared by adding 360 gm of glucose to 864 g of water. Calculate mole fraction of glucose (molar mass of glucose = 180)

$\Rightarrow$ $n_{glucose} = \frac{360}{180} = 2$ mole

$n_{water} = \frac{864}{18} = 48$ mol.

$$X_g = \frac{n_g}{n_g + n_w} = \frac{2}{2+48} = \frac{1}{25}$$

Molarity

no of moles of solute dissolved in 1 Litre soln.

$$\text{Molarity } (M) = \frac{\text{no of moles of solute}}{\text{Vol of solution in Litre}}$$

$$M = \frac{\text{weight of solute}}{\text{molar mass of solute} \times \text{Vol of soln}^{n}(l)}$$

Unit = mol/Litre or M (Molar)

* To Cal, molarity of a solution after dilution :-

$$M_1V_1 = M_2V_2$$

M_1 — initial molarity
V_1 — initial vol
M_2 — final molarity
V_2 — final vol.

* For mixing two solution of same substance

$$M_1V_1 + M_2V_2 = M_R(V_1 + V_2)$$

$$M_R = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

* $$\text{Molarity} = \frac{\%\text{ by weight} \times 10 \times d}{GMM}$$ (d = density)

Qu A given soln of NaOH contains 4 gram NaOH Per Litre of Solution. Cal Molarity.

$$\Rightarrow n_{NaOH} = \frac{4}{40} \qquad M = \frac{\frac{1}{10}}{1 \text{ Litre}} = 0.1M \text{ Ans}$$

Qu A solution is prepared by dissolving 1 gram of NaOH in water to get 250 ml of solution. Cal molarity.

$$n_{NaOH} = \frac{1}{40} \qquad M = \frac{\frac{1}{40}}{\frac{250}{1000}} = \frac{1}{40} \times \frac{1000}{250} = \frac{1}{10} M$$

Qu How many moles & how many grams of HCl are present in 250 cc of 0.5 M HCl.

(1 cc = 1 ml)

$\Rightarrow$ $M = 0.5$

$$M = \frac{n}{V(l)}$$

$$0.5 = \frac{n}{250} \times 1000^{4}$$

$$4n = 0.5$$

$$n = \frac{0.5}{4}$$

$$\boxed{n = 0.125}$$

mass

$= 0.125 \times 36.5$

$= 4.5625$ gm.

Molality

$$\text{Molality } (m) = \frac{\text{no of moles of Solute}}{\text{Mass of solvent in kg}}$$

Unit = mol/kg or molal (m)

Qu a solution contains 10 moles of sucrose in 1 kg of solvent. Calculate molality.

$$m = \frac{10}{1} = 10$$

Qu Cal molality of a solution containing 5.3 gm of anhydrous Na_2CO_3 in 400 g water. ($M_{Na_2CO_3} = 106$)

$$m = \frac{5.3/106}{400/1000}$$

$$= \frac{0.05}{0.4}$$

$$= 0.125\ m$$

Qu. 225 gm of aq soln contains 5 gram urea. Find molality. (molar mass of urea = 60)

⇒ mass of urea = 5g

Solution = 225gm

Solvent = 220 gm

$$m = \frac{5/60}{220/1000}$$

$$m = \frac{5}{3} \times \frac{5}{220} = \frac{25}{66}$$

$$= 0.37$$